

**RESEARCH & INSIGHTS HUB**

Bitcoin Halving and Miner Economics

Following the halving, miner revenue per terahash is set to drop to all-time-lows absent significant bitcoin price appreciation, and transaction fee based revenues are growing in importance.

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Bitcoin's exponential hash rate growth to date underpins the strength of bitcoin's economic security. An increase in the hash rate is an increase in the cost to fork or otherwise reorganize the network. In our view, however, hash rate growth is not limitless and may increasingly slow – or even shrink – following subsequent halving cycles.

Mining machines have had smaller proportional efficiency gains relative to the growth of the network hash rate, and without significant bitcoin price appreciation, the revenue per unit of hash power could drop to all time lows following the halving in April 2024. Thinner margins could result in a larger proportion of forced selling by miners in order to meet fixed operational and financing costs, possibly making some miners unprofitable and increasing consolidation by well-capitalized entities who are able to acquire distressed assets at a discount.

Although this is not an imminent threat to Bitcoin's sustainability, the drop in fixed block rewards does underscore the need for innovation that can drive miner revenue to support a strong network hash rate. In fact, recent surges in transaction fees have had an outsized impact on miner revenues compared to price movements. Absent any drastic price shocks, however, our overall view is that the network hash rate will likely continue to grow post-halving in the short-to-medium term,

but may subsequently stall absent price appreciation or increased transaction fees.

For more details, see our [full report](#).

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